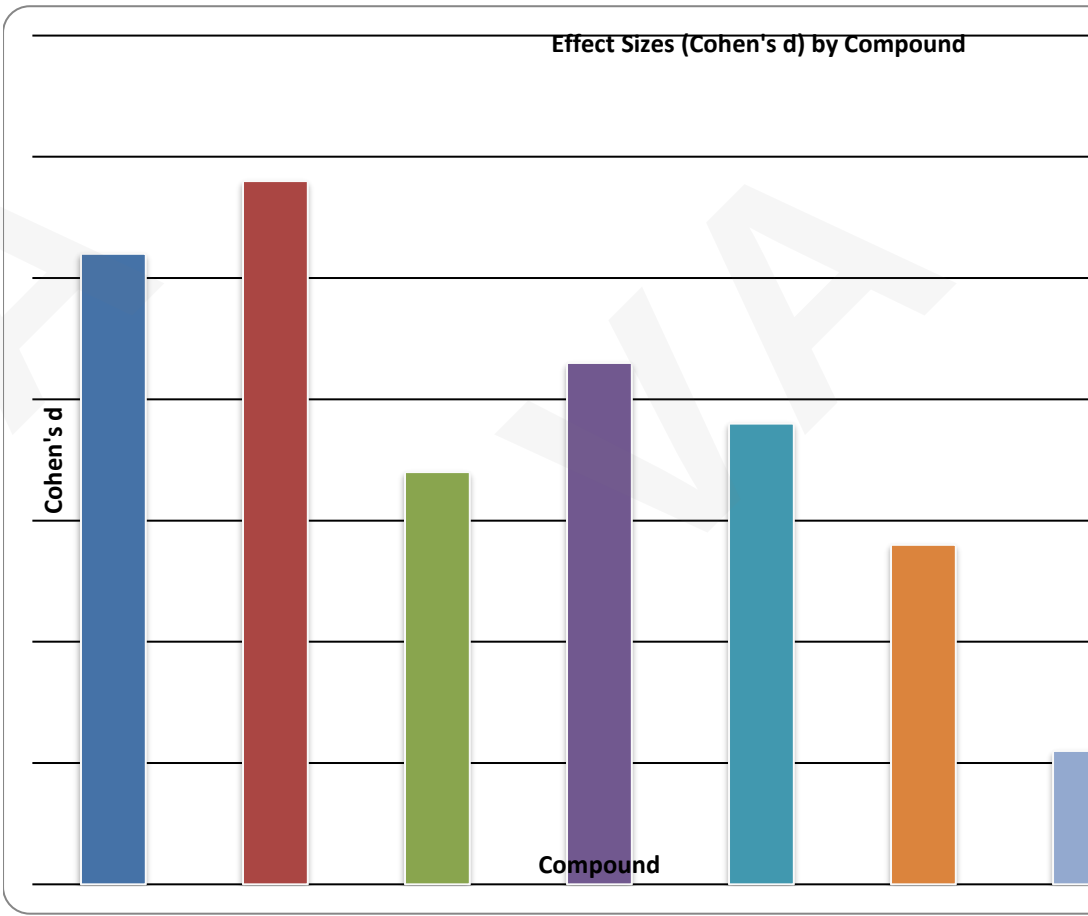


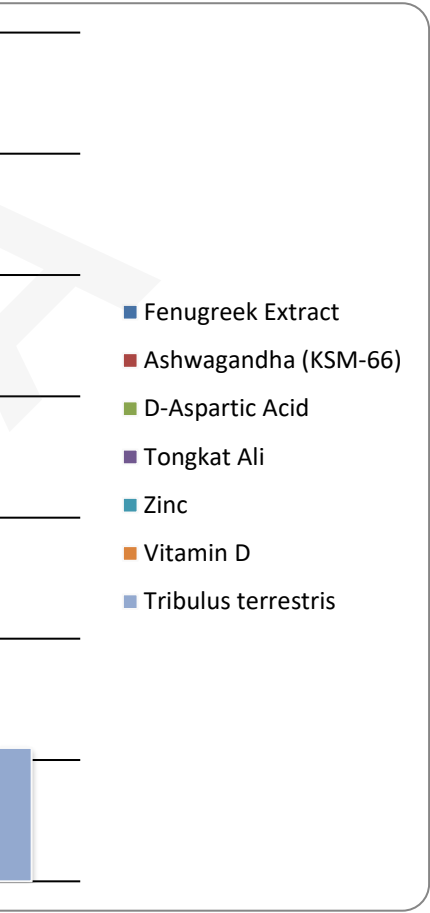
Table 1. Meta-Analytic Summary

Compound	Studies (n)	Total N	Typical Dose
Fenugreek Extract	8	524	500 mg/day
Ashwagandha (KSM-66)	7	438	600 mg/day
D-Aspartic Acid	6	312	3 g/day
Tongkat Ali	5	287	200 mg/day
Zinc	6	341	30 mg/day
Vitamin D	5	298	3,332 IU/day
Tribulus terrestris	4	196	750 mg/day
POOLED (All Compounds)	41	2396	—



ary of Natural Testosterone Compound Efficacy

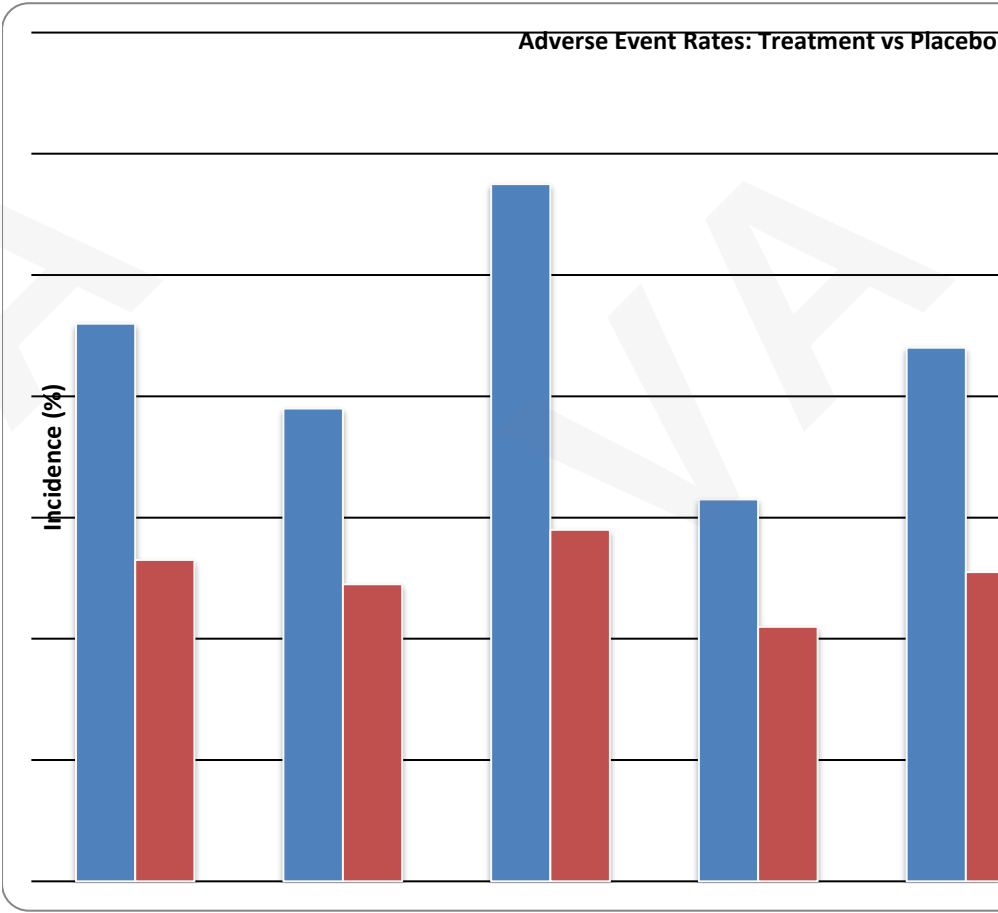
Cohen's d	CI Lower	CI Upper	I ² (%)
0.52	0.31	0.73	42
0.58	0.34	0.82	38
0.34	0.12	0.56	55
0.43	0.18	0.68	47
0.38	0.15	0.61	51
0.28	0.06	0.5	44
0.11	-0.09	0.31	31
0.41	0.28	0.54	46



p-value
0.001
<0.001
0.003
0.002
0.004
0.014
0.287
<0.001

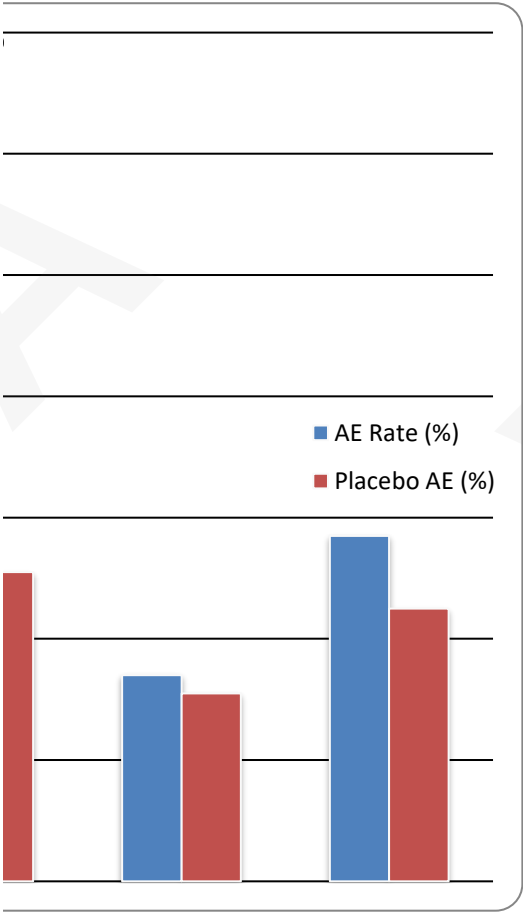
Table 2. Adverse Event I

Compound	AE Rate (%)	Placebo AE (%)
Fenugreek	9.2	5.3
Ashwagandha	7.8	4.9
D-Aspartic Acid	11.5	5.8
Tongkat Ali	6.3	4.2
Zinc	8.8	5.1
Vitamin D	3.4	3.1
Tribulus terrestris	5.7	4.5
POOLED	7.528571429	4.7



Profiles for Natural Testosterone Compounds

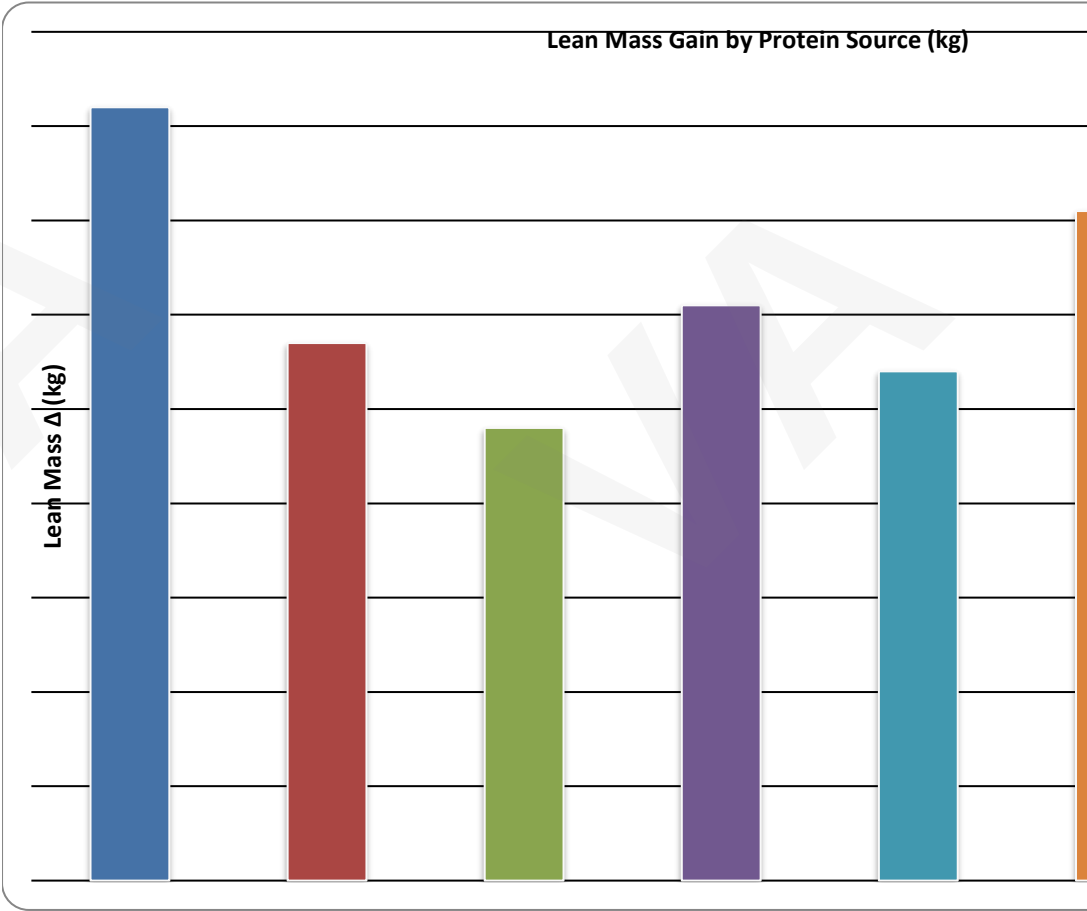
Risk Difference (%)	Common AEs	Serious AEs
3.9	comfort, maple syrup body	0
2.9	Mild drowsiness, GI upset	0
5.7	headache, irritability, bloating	0
2.1	Restlessness, mild insomnia	0
3.7	a, metallic taste (at higher c	0
0.3	None frequently reported	0
1.2	GI discomfort	0
2.828571429	dominantly mild GI sympto	0



Dropout Rate (%)
3.1
2.4
4.8
2.1
1.9
1.2
2.6
2.585714286

Table 3. Protein Supplementation Efficacy

Source	Studies (n)	Total N	Lean Mass Δ (kg)
Whey Protein	14	687	0.82
Casein	5	214	0.57
Soy Protein	4	186	0.48
Pea Protein	4	178	0.61
Rice Protein	3	132	0.54
Blended (Pea+Rice)	3	148	0.71
POOLED	33	1545	0.69



Protein Quality by Source (8-12 Week Interventions)

CI Lower	CI Upper	Strength Δ (%)	DIAAS Score
0.54	1.1	8.2	1.09
0.21	0.93	5.8	1
0.12	0.84	4.9	0.9
0.28	0.94	6.7	0.82
0.14	0.94	5.4	0.37
0.33	1.09	7.3	0.87
0.47	0.91	6.383333333	—

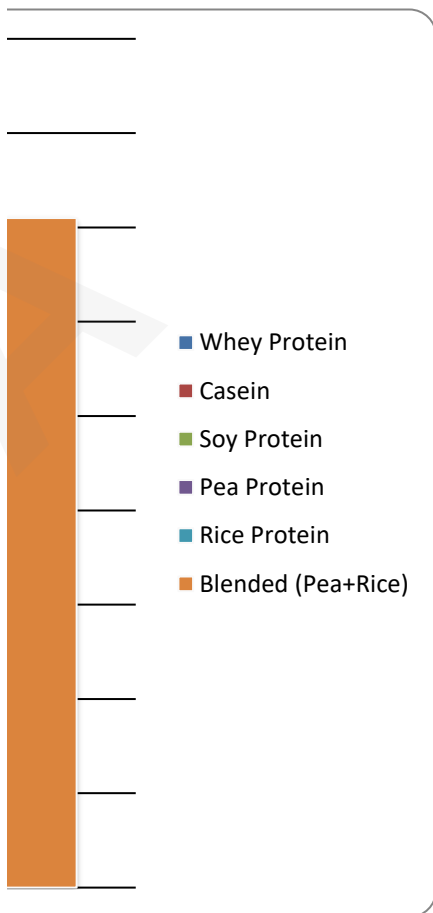


Table 4. Side-Effect Incidence by Pro

Daily Intake (g/kg)	GI Symptoms (%)	Renal Biomarker Shift	GFR Elevation (%)
< 1.6	3.2	No change	0
1.6 - 2.2	5.8	No significant change	0.8
2.2 - 3.0	11.4	Mild GFR elevation	4.2
> 3.0	18.7	GFR elevation	8.9

Protein Dose Range

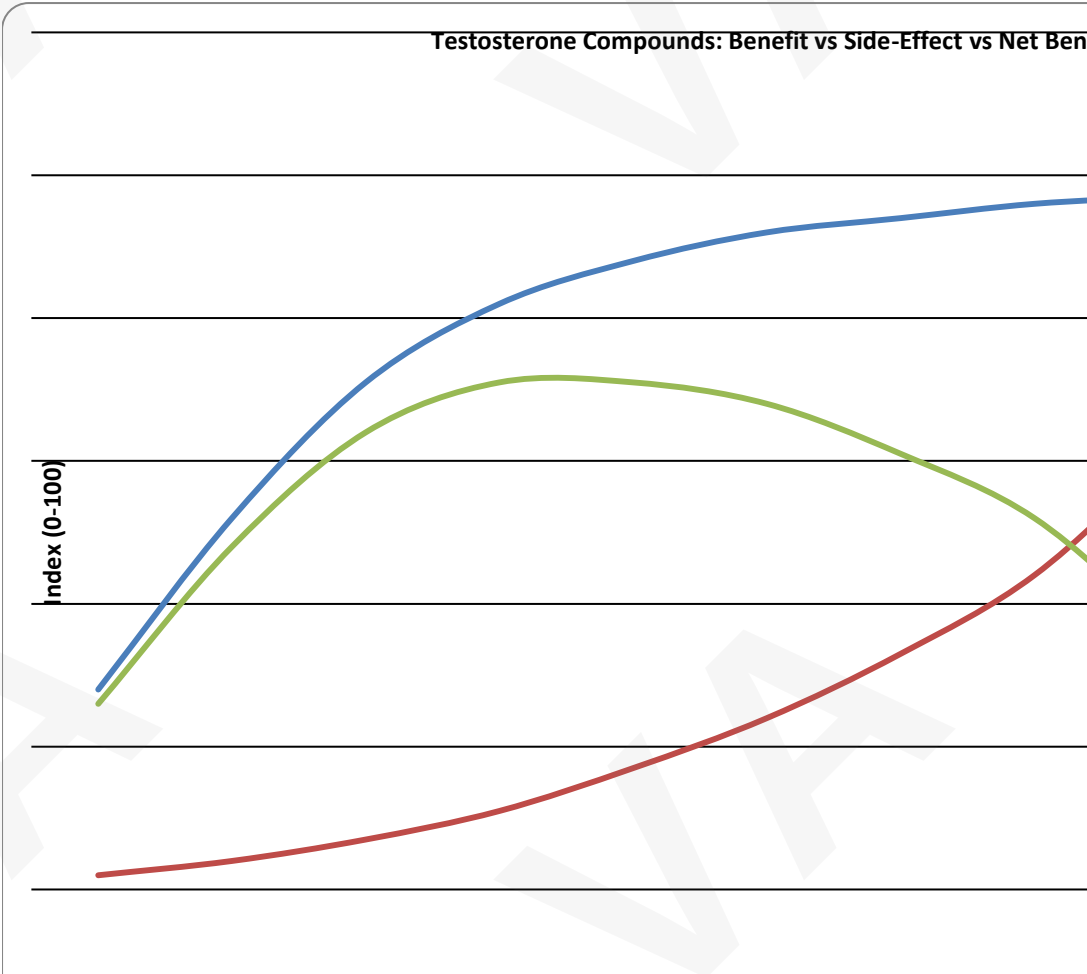
Hepatic Marker Shift	ALT Rise (%)	Total AE (%)
No change	0	3.2
No change	0	6.6
No change	0	15.6
Minor ALT rise	2.3	29.9

Dose-Response Data: Benefit Index vs Side-Ef

Panel A: Natural Testosterone Compounds (Normalised Dose-Response)

Dose Multiplier (x RDA)	Benefit Index (0-100)	Side-Effect Index (0-100)	Net Benefit Index
0.25	28	2	26
0.5	52	4	48
0.75	71	7	64
1	82	11	71
1.25	88	17	71
1.5	92	24	68
1.75	94	33	61
2	96	44	52
2.5	97	61	36
3	98	79	19
3.5	98	92	6
4	99	100	-1

Peak NBI at dose: 1
Peak NBI value: 71
Zero-crossing dose: ~3.5x



Dose Multiplier (x Recommended)

Effect Index

Benefit Slope	Harm Slope
28	2
24	2
19	3
11	4
6	6
4	7
2	9
2	11
1	17
1	18
0	13
1	8

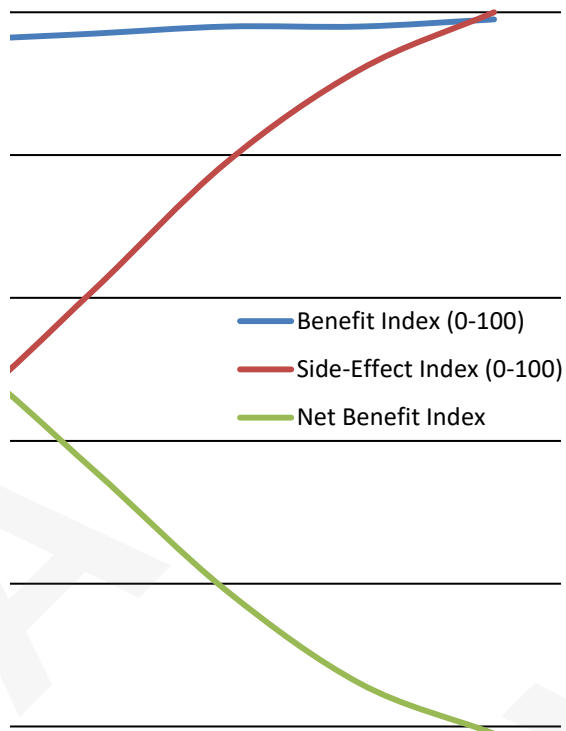
Panel B: Protein Sup

Protein (g/kg/day)
0.8
1
1.2
1.4
1.6
1.8
2
2.2
2.5
3
3.5
4

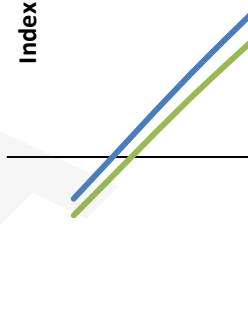
Peak NBI at dose:

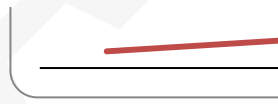
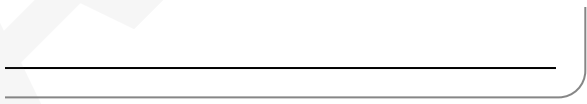
Peak NBI value:

Benefit (by Dose)



Index (0-100)



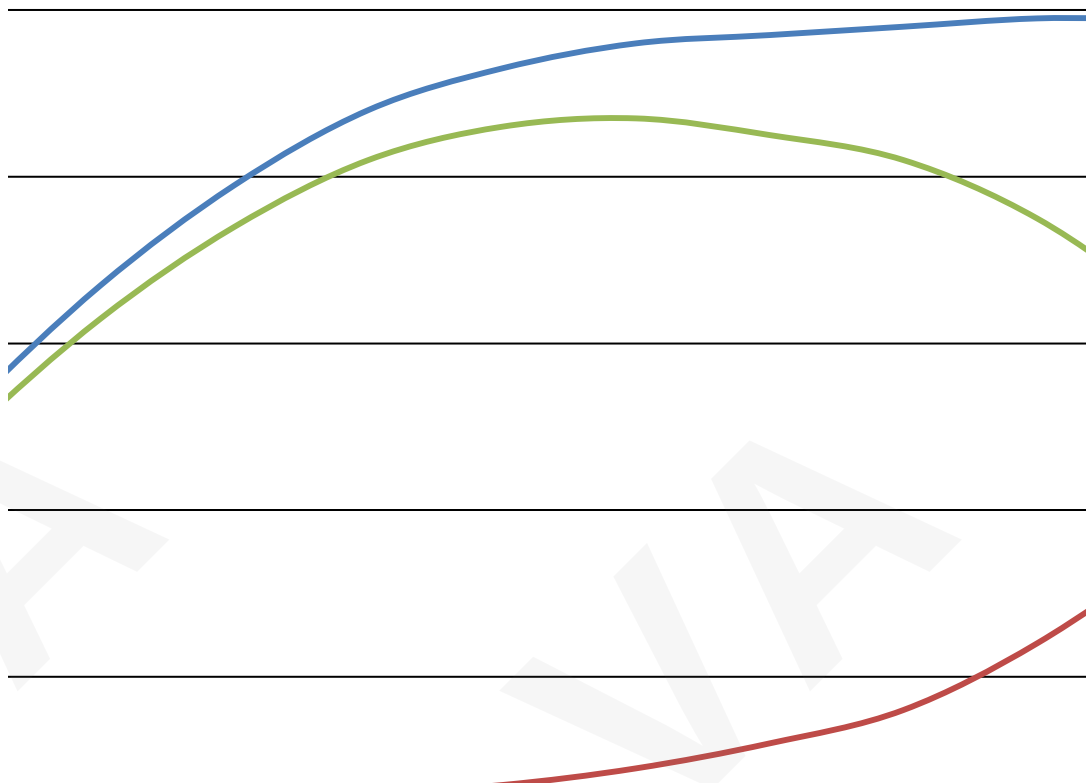


plementation (Dose-Response by g/kg/day)

Benefit Index	Side-Effect Index	Net Benefit Index	MPS Plateau?
35	2	33	No
52	3	49	No
67	4	63	No
79	5	74	Approaching
88	6	82	Near plateau
93	7	86	At plateau
96	9	87	Past plateau
97	12	85	Past plateau
98	16	82	Past plateau
99	24	75	Past plateau
99	35	64	Past plateau
100	48	52	Past plateau

2
87

Protein: Benefit vs Side-Effect vs Net Benefit (by Dose)



Protein Intake (g/kg/day)

Risk Level
Minimal
Minimal
Low
Low
Low
Low
Low-Moderate
Moderate
Moderate
Moderate-High
High
High

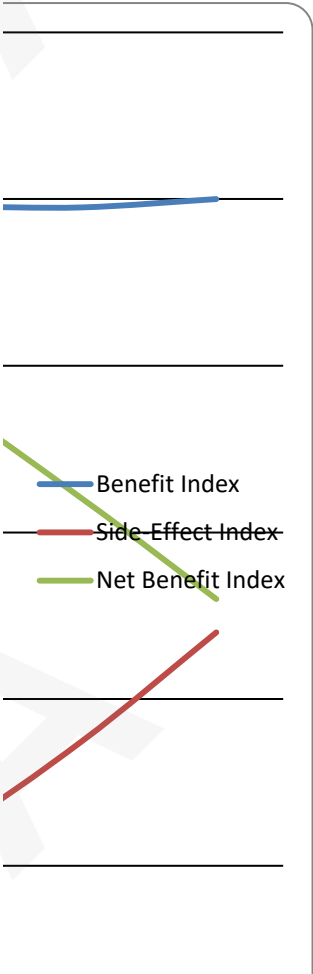
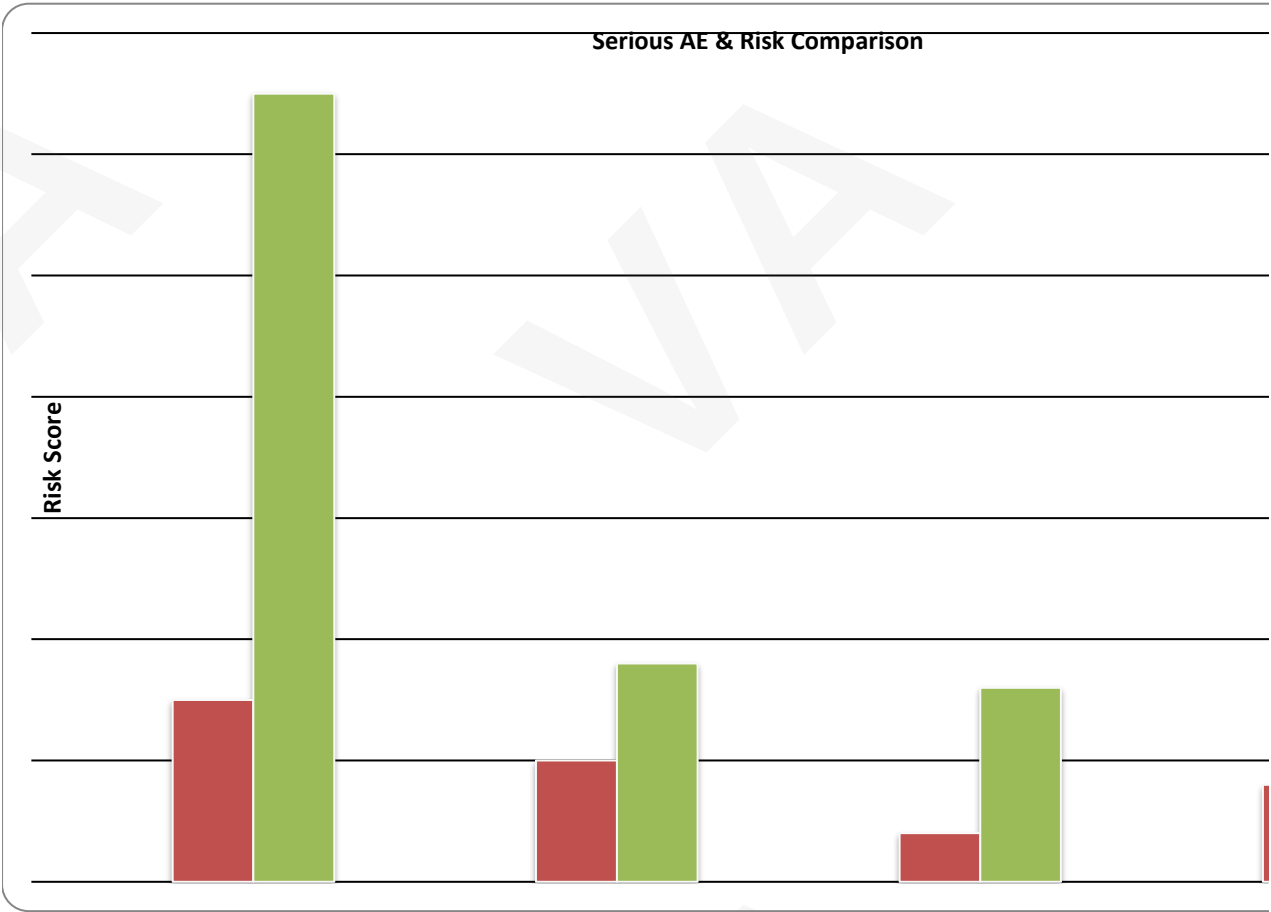




Table 5. Natural vs Synthetic Testosterone: Cor

Metric	Natural Compounds	TRT (Clinical Dose)
Testosterone Δ (%) - Low	15	200
Testosterone Δ (%) - High	40	400
Lean Mass Δ (kg/12wk) - Low	0.5	3
Lean Mass Δ (kg/12wk) - High	1.2	6
Mild AE Incidence (%) - Low	6	25
Mild AE Incidence (%) - High	12	40
Serious AE Incidence (%)	0	7.5
HPG Axis Suppression (severity 0-1	0	5
Hepatotoxicity Risk (0-10)	0	2
Cardiovascular Risk (0-10)	0	4



Comparative Summary

AAS (Recreational)	Natural as % of AAS
500	3.0
3000	1.3
5	10.0
15	8.0
60	10.0
85	14.1
32.5	0.0
9	0.0
8	0.0
8	0.0

